

G8-DESIGN-AUDIT-v1: Mapping bridges to $\int_0^{T^*} W_+ < \infty$

ZFL-CX

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1 Frozen input G7

$$\boxed{G7 = \text{FROZEN}} \quad \boxed{\mathcal{C}_{\text{PROVEN}} = \{C_E^{\text{weak}}, C_{L^3}, C_F^{\text{conc}}, C_I^{\text{freq}}, C_J^{\text{neg}}\}} \quad \boxed{\mathcal{C}_{\text{PROVEN}} \not\Rightarrow \perp \quad \text{OPEN}}$$

Bottleneck: $\boxed{T^* < \infty \Rightarrow \int_0^{T^*} W_+(t) dt = +\infty}, W(t) = \int \omega^T S \omega.$

2 Rule for G8-01 creation

No G8-01 until: hypothesis + mechanism + quantitative target

$$\boxed{\text{Target A: } \int_0^{T^*} W_+ < \infty \text{ or Target B: } \mathcal{C}_{\text{PROVEN}} \wedge C_X \Rightarrow \perp}$$

3 Candidate audit

3.1 1. Algebraic stretching structure

Attempt: $\boxed{W_+(t) \leq \Phi(C_F, C_I, C_J)}$ with $\int_0^{T^*} \Phi < \infty$. Form $W = \int \omega^T S \omega$. Known $|W| \lesssim \|u\|_3 \|\nabla \omega\|_2^2$ fails because C_{L^3} allows blow-up. Status: $\boxed{\text{OPEN}}$ — only if new inequality not reducible to $\|u\|_3$. Kill condition: if $\Phi \sim \|u\|_3^a$ or $f = +\infty$ possible, $\Rightarrow$ KILLED.

3.2 2. Critical scale + concentration geometry

Not “does concentration exist” — closed. New: $\boxed{\text{concentration must have obligatory geometry limiting stretching?}}$
 Need: incompatibility conc. L^3 vs positive W production. Status: $\boxed{\text{BLOCKED}}$ — $C_F^{\text{conc}} + C_I^{\text{freq}}$ already same phenomenon $r_n \sim N_n^{-1}$, no quantitative W_+ bound found in G7. Kill condition: only $r_n \rightarrow 0$ without structure $\Rightarrow$ KILLED.

3.3 3. Pressure-strain nonlocal

$D_t S = -S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S, p = \Delta^{-1} \partial_i \partial_j (u_i u_j)$. Question: $\boxed{\text{Does nonlocal } p \text{ impose global restriction on } (\omega^T S \omega)}$
 Need: $W_+ \leq F$ (controlled) or geometric incompatibility, not just $\|\nabla^2 p\|_q \lesssim \|S^2 - \Omega^2\|_q$. Status: $\boxed{\text{VIALE/ OPEN}}$ — highest audit priority, not tested in G7 as independent bridge. Kill condition: if only Calderon-Zygmund, $\Rightarrow$ KILLED.

3.4 4. New critical functional C_X

Need: $C_X : T^* < \infty \Rightarrow X(t) \rightarrow \infty$ critical, and $C_X \wedge \mathcal{C}_{\text{PROVEN}} \Rightarrow \perp$ or $C_X \Rightarrow \int W_+ < \infty$. Not

“BMO⁻¹couldhelp?” Status : OPEN || requires concrete X with mechanism, not generic embedding. Kill condition : nobridge to W_+ or reproduces C_{L^3} .

4 Priority order

stretching algebraic $\rightarrow$ pressure-strain $\rightarrow$ critical functional $\rightarrow$ geometric concentration

No CKN/frequency/concentration pure restart.

5 Decision

G8-01 NOT AUTHORIZED — audit only

Next: choose ONE viable with concrete Φ or C_X